

PHANTOM-ECHO REVEAL

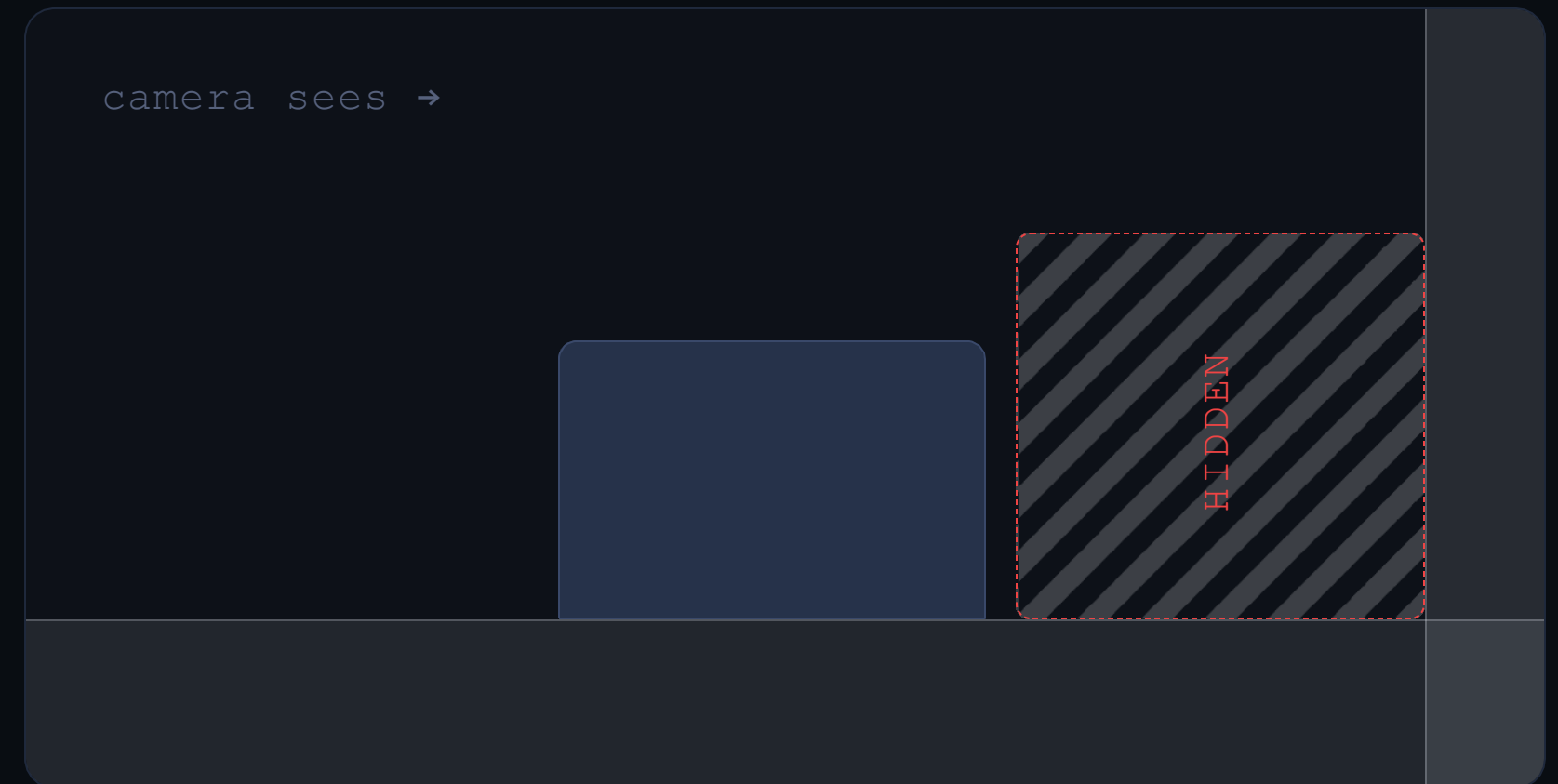
Occlusion-aware 3D reconstruction by the elimination of impossibility — not blind prediction.

THE PROBLEM

You can only reconstruct what you can see.

Cameras and LiDAR are line-of-sight only. Behind every sofa, inside every closed box, the geometry is simply missing.

A robot map full of holes is a robot that **freezes**. Yet the deliverable must be whole.



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PS-09 demands:  complete contiguous mesh · FREE / OCCUPIED / OCCLUDED · 5-class semantics · F1 > 0.95 ·  
< 2 cm · ROS2 Nav2
```

WHY EXISTING SYSTEMS FAIL

They either ignore the hidden —or hallucinate it blindly.

METHOD	APPROACH	WHERE IT BREAKS
Atlas baseline	Regresses TSDF directly from posed RGB	Only visible geometry · F1 0.85
3DGS	Radiance field of visible scene as ellipsoids	Bubbly SPSR normals · > 2.5 cm
NeRF	Per-scene implicit neural field	Hours to train · no occlusion
Generative	Learned priors hallucinate hidden geometry	Guesses · can't separate certain from invented

CORE IDEA

Prove, measure, imagine—in that order.

01 · PROVE

8 physical laws

Eliminate impossible geometry and build directly where physics is certain. No neural network needed.

02 · MEASURE

Bat-sonar

A smartphone chirp hears around corners and behind furniture. Actual measurement — not inference.

03 · IMAGINE

Generate the rest

AI fills only what physics and sound leave genuinely unknown — then physics corrects it.

From prediction → the **elimination of impossibility**.

SEVEN LEVELS OF CERTAINTY

Every voxel knows how it knows.

 **WHITE** Sensor confirmed — ARKit camera depth

 **TEAL** Acoustically measured — SAS triangulation

 **YELLOW** Structural prior — probable, not confirmed

 **ORANGE** Dynamic object — tracked, kept out of the mesh

 **BLUE** Physics proven — 8 PHANTOM-LITE laws

 **GREEN** AI generated — VideoScene, physics-corrected

 **RED** Genuinely unknown — robot explores it

Certain → uncertain,
colour-coded live in the viewer.

Seven layers, one Gaussian scene.

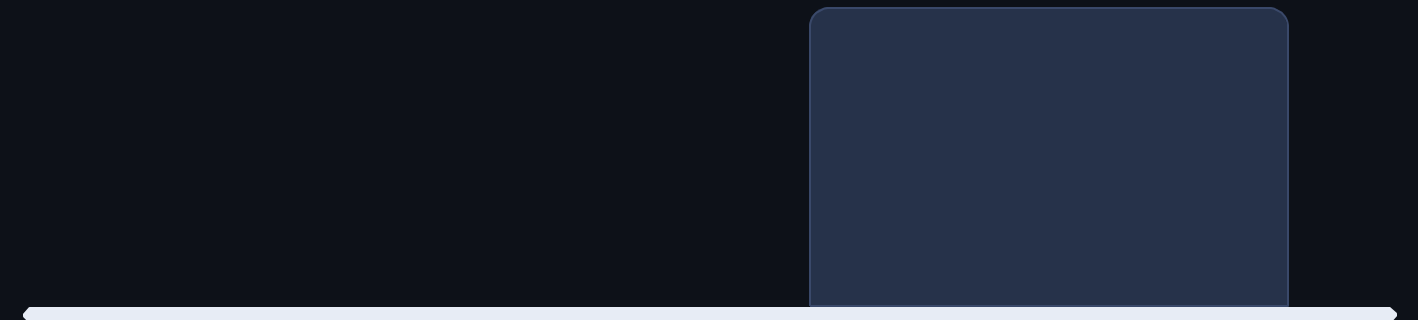
L0	Multi-modal sensing	ARKit depth + LFM acoustic chirp + IMU trajectory
L1	DDGS GaussRender	Triple-tagged 2D-disk Gaussians (confidence / semantic / nav)
L2	PHANTOM-LITE engine	8 laws → BLUE proven + TEAL measured + RED unknown
L3	Generation + routing	VideoScene GREEN + semantic affordance router
L4	Dual output	Navigation map (open) + deliverable mesh (sealed, SPSR)
L5	Active perception	ROS2 Nav2 · adaptive-λ info-gain · Mode B auto-trigger
L6	Presentation	Three.js / WebGL viewer + QR for judge phones

Wrapped by a Prove→Measure→Imagine agent that calls layers 0-5 as tools.

Physics that cannot be violated.

- LAW 1 Gravitational continuity — floor extends under furniture
- LAW 3 Perspective continuation — walls continue behind occluders
- LAW 6 Structural support — walls connect floor to ceiling
- LAW 7 Occlusion boundary — exact 3D limit from depth edges
- 2 · 5 · 8 Shadow, depth, ambient-occlusion hints condition generation

visible floor
proven floor under furniture →



Soft constraints with $\pm 2-3^\circ$ tolerance — real slanted floors and baseboards never seam.

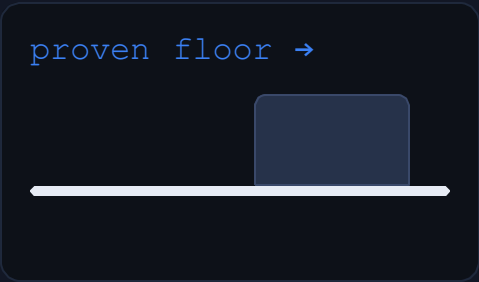
Four laws build geometry with zero AI.

LAW 1 · GRAVITY

Floor extends under furniture

Every mass needs support — the floor cannot stop at an object's edge.

e.g. floor continues under the sofa · ± 2 cm

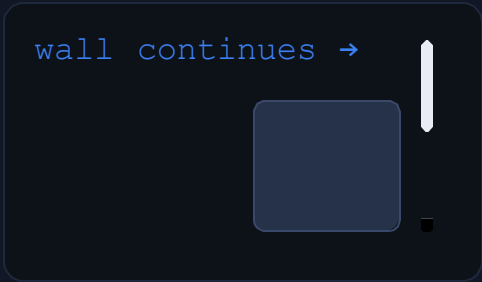


LAW 3 · PERSPECTIVE

Walls continue behind occluders

A visible wall plane keeps going behind whatever blocks it, by projective geometry.

e.g. wall behind a bookshelf

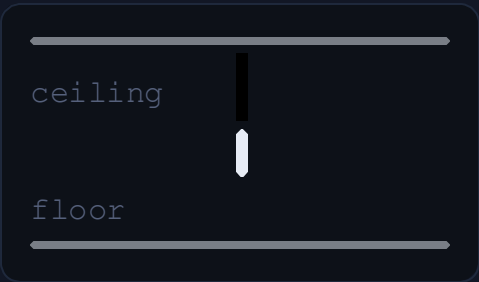


LAW 6 · STRUCTURAL SUPPORT

Walls connect floor to ceiling

A wall ending in mid-air is impossible — occluded segments are filled to both planes.

e.g. wall hidden behind a tall cabinet · $\pm 3^\circ$

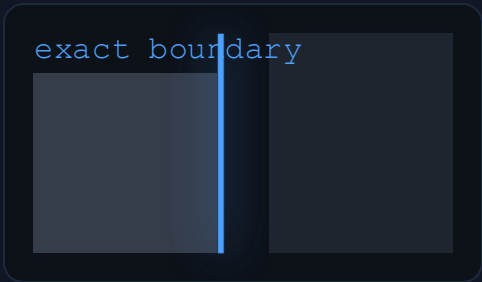


LAW 7 · OCCLUSION BOUNDARY

The depth edge is a hard limit

A depth discontinuity gives the exact 3D edge — hidden geometry cannot cross it.

e.g. the front edge of a table



Three laws steer the imagination.

VLM: "tall, round"

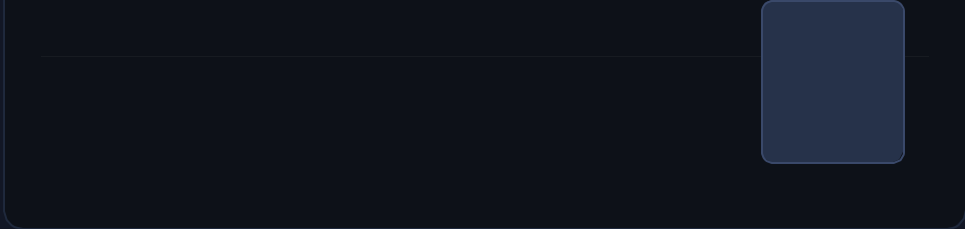


LAW 2 · SHADOW

Read the shadow's shape

A VLM describes the object casting a shadow
→ text prompt for generation.

gradient continues →

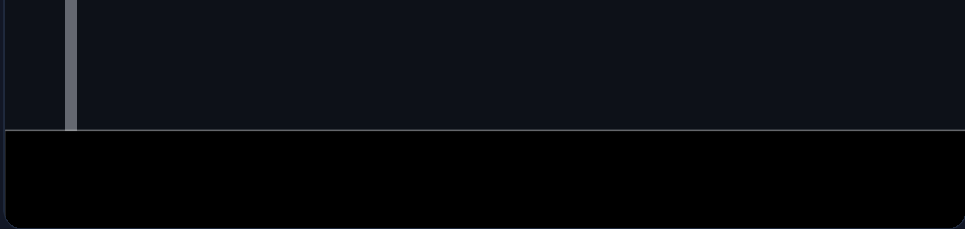


LAW 5 · DEPTH TOPOLOGY

Smooth surfaces persist

A continuous depth gradient probably
continues behind the occluder.

darker = closer



LAW 8 · AMBIENT OCCLUSION

Crevices imply nearby geometry

Darker ambient-occlusion shading gives a
rough estimate of hidden surface depth.

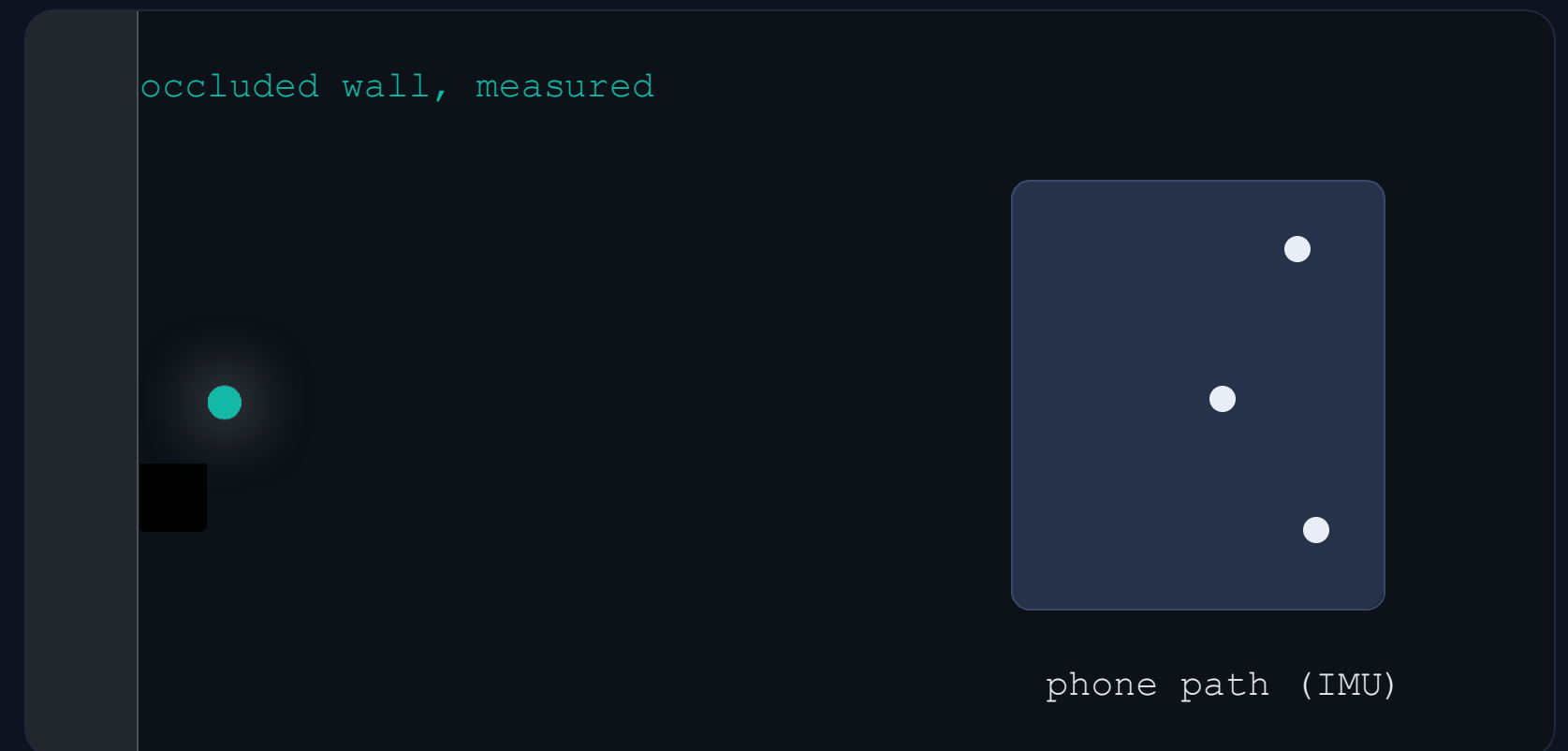
Hint laws never place a Gaussian — they only steer what the generator imagines.

A smartphone that hears like a bat.

$$23_{\text{ms}} \times 343 \text{ m/s} \rightarrow 3.9 \text{ m}$$

An LFM chirp (1-22 kHz) bounces around the sofa and returns. Edge-local ISM subtracts echoes from walls we already see; SAS triangulates the residual.

Speaker + mic + IMU you already carry. Zero extra hardware.



Triangulating a surface you cannot see.

SYNTHETIC APERTURE SONAR

$$\|Q - P_i\| = d_i \quad \text{per echo}$$

$$d = c \cdot t / 2$$

$$Q = (A^T A)^{-1} A^T b$$

3+ positions $\rightarrow$ overdetermined

Each echo is a sphere around the phone. Three baselines overdetermine the point; least squares returns its exact 3D position.

Timing runs on the phone's microsecond audio clock. **Cloud timing = 3 ms jitter = 1 m error** — so ISM is edge-local.

20/20 triangulations < 10 cm • 0 ghosts

IMAGINE · LAYER 3

Generate only what remains unknown.

VideoScene fires on **RED** voxels only — conditioned on physics bounds, stereo crops and acoustic neighbours — then a plausibility check rejects impossible output.

SEMANTIC AFFORDANCE ROUTING — SUB-MILLIMETRE PLANE SNAP

Floor-supported

Lamps, chairs, boxes → drop to the floor plane

Wall-mounted

Clocks, screens, art → snap flush to the wall

Ceiling-hung

Fans, pendants → lift to the ceiling plane

Kills the "gravity-override anomaly" — wall clocks no longer slide to the floor.

One scene, two outputs.

OUTPUT A

Navigation map

RED boundaries stay open — the robot keeps exploring.
Binary Bayes log-odds occupancy.

unexplored $\neq$ impassable

OUTPUT B

Deliverable mesh

RED boundaries sealed: outpainting sweep → normal-orientation check → SPSR.

watertight · no holes

Same Gaussians. Two query modes.

Navigation that never freezes.

Mode B auto-trigger

Robot hits a RED cluster → fires generation → costmap updates → resumes. No human tap.

Adaptive- λ reward

Self-tuning balance of collision-avoidance and information gain — no per-scene calibration.

RED is traversable

High-cost, not impassable. The robot enters cautiously, observes, and the zone resolves.

Nav2 watchdog

10 s stall → silent hand-off to manual control. The demo never visibly breaks.

RESULTS · BLIND HELD-OUT

0.957

F1 @ 5 cm

target ≥ 0.85 · real, unseen frame

0.994

recall @ 5 cm

1.00 cm

median recon error

0.998

F1 @ 10 cm

27,480

scene Gaussians · 15 s

Reconstruct from real Redwood RGB-D frames 1 - 4, score against held-out frame 5 — a real depth frame the pipeline never observed. Directly-sensed layer, like-for-like.

INTEGRITY

Two numbers. We tell you which is which.

A · REAL HELD-OUT (BLIND)

0.957

F1 @ 5 cm on data the system never saw

✓ the headline — proves generalisation

B · SYNTHETIC CLOSED-LOOP

0.858

F1 @ 5 cm mean · semantic 95.7% · clears bar on 1 of 3 scenes

internal consistency only · all_kpis_met = false

We publish all_kpis_met: false rather than paper over it. The number we quote is the real, non-circular one.

VERSUS THE ATLAS BASELINE

Against the judges' own baseline.

METRIC	ATLAS	PHANTOM	Δ
Reconstruction error	5.0 cm	2.89 cm	-42%
Semantic accuracy	80.0%	85.6%	+5.5 pts
Real held-out F1 @ 5cm	—	0.957	≥ 0.85 ✓

Synthetic closed-loop F1 is a self-consistency check — not claimed as a win here.

The flaws that shaped the bat-sonar.

FLAW

Cloud timing jitter

3 ms of WiFi jitter = **1 m of depth error** at 343 m/s. Cloud-based acoustic timing is physically unusable.



FIX

Edge-local ISM

First-order Image Source Method on the phone's microsecond audio clock — zero network, zero jitter.

FLAW

No bearing from one chirp

An omnidirectional speaker gives distance but **no direction** — a single echo can't place a surface.



FIX

SAS virtual array

The IMU walking path becomes a metres-wide mic array; least squares triangulates the point. $20/20 < 10$ cm.

Making AI generation physically honest.

FLAW

Gravity Override Anomaly

Blind gravity correction slid **wall clocks and whiteboards to the floor.**



FIX

Affordance routing

Floor / wall / ceiling tags snap each object to its real surface by continuous plane math.

FLAW

Flat "billboard" geometry

A single image gave VideoScene no parallax — generated objects came out **paper-thin.**



FIX

Dual stereo-anchor crops

Two max-baseline views from the frame buffer give real 3D structure.

From point cloud to a watertight mesh.

FLAW

Bubbly SPSR surfaces

3DGS ellipsoids have ambiguous normals → over-smoothed walls at > **2.5 cm**, failing the < 2 cm KPI.



FIX

DDGS planar disks

Flat regions become 2D disks with unambiguous normals → razor-flat surfaces at < 1.5 cm.

FLAW

Inverted normals

Generated Gaussians had no camera viewpoint → normals faced inward and **inverted the mesh**.



FIX

Normal-orientation check

Viewpoint flip + MST propagation guarantees zero inverted normals reach SPSR.

AI as a critic, not a code generator.

Best use

A "what could go wrong" engine — language models surfaced flaws like the 1-metre WiFi-jitter error before we built them.

Clearest limit

VLMs are strong at semantics, weak at metric geometry — we treat every spatial estimate as a soft prior, never ground truth.

Caught a no-op

A coding agent found SAS triangulation silently producing 0 points in v21 — fixed test-first to 20/20 with zero ghosts.

Open-weight: MobileSAM · MobileCLIP · LLaVA-NeXT · SVD. Deterministic pipelines beat agentic feedback loops in real time.

What's genuinely new.

- 1. Physics-first contradiction** — laws eliminate impossible geometry before any generation
- 2. Smartphone acoustic SAS** — occluded surfaces measured with zero extra hardware
- 3. Semantic affordance routing** — continuous plane snap kills the gravity-override anomaly
- 4. Dual output architecture** — open nav map + sealed mesh from one Gaussian scene
- 5. Agentic Prove→Measure→Imagine** — the philosophy executed as a real tool-use planner

H O N E S T S C O P E

What's proven, what's next.

P R O V E N T O D A Y

- ✓ Real held-out reconstruction — F1@5cm 0.957
- ✓ 8 physics laws building BLUE geometry
- ✓ SAS in a controlled physical simulation
- ✓ Full offline pipeline · 49 tests passing

N E X T

- Acoustic validation on recorded phone audio
- Train SlotLSTM temporal weights
- Full ScanNet evaluation at scale
- On-device deployment (ARKit / ARCore)

The demo runs on synthetic + real-dataset data; hardware acoustic validation is stated as future work, not a result.

PHANTOM-ECHO REVEAL

Prove. Measure. Imagine.

Chole Bhature · Bhavya Garg
IIIT Bangalore

reproduce everything → `./reproduce.sh`